Assignment 8: DH Parameters

Robot Kinematics and Dynamics

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Due: See Canvas for the due date

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1 Overview

This assignment reinforces the following topics:

- DH Parameters

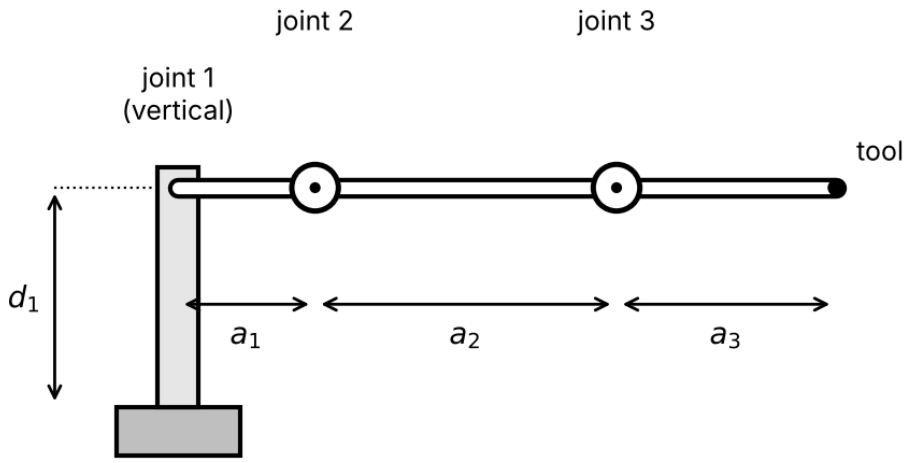
2 Written Questions

For all of the following questions, please fully compute all answers and show your work for full credit, unless otherwise specified. Submit your answers in a PDF entitled `writeup.pdf` in your handin directory. All answers must be typed, but diagrams may be hand-drawn and scanned in. However, they must be tidy and fully legible! Consider drawing them in a black or blue pen. All units are in radians and meters, where appropriate.

1) DH Frames and Parameters

The RRR arm below is shown at its zero pose ($\theta_1 = \theta_2 = \theta_3 = 0$). Joint 1 turns about the vertical axis, and joints 2 and 3 turn about axes perpendicular to the page. Place the base frame (frame 0) at the bottom of the vertical column with z_0 up, x_0 to the right and y_0 into the page. Frame 3 is the tool frame at the end of link 3. A positive rotation of joints 2 and 3 raises the arm.

Zero pose ($\theta_1 = \theta_2 = \theta_3 = 0$). Joints 2 and 3 turn about axes perpendicular to the page; joint 1 about the vertical.



(1) [4 points] Using the Denavit-Hartenberg convention, state the direction (x_i, y_i, z_i) of frames 1, 2 and 3 at the zero pose, and where each origin is. A sketch on the figure is fine.

(2) [4 points] Fill in the DH table (a_i, α_i, d_i, θ_i) for the arm, marking the joint variables.

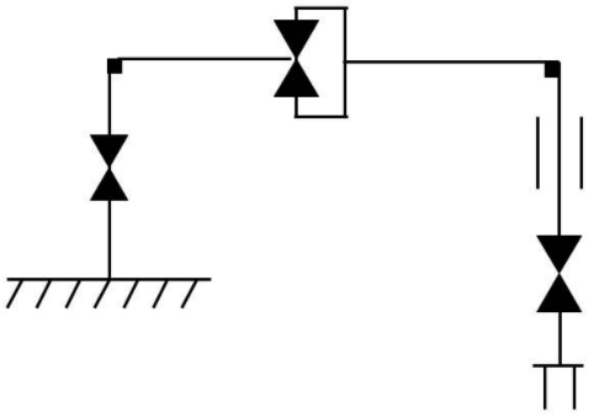
(3) [6 points] Using your table, find the position of the origin of frame 3 in the base frame as a function of $\theta_1, \theta_2, \theta_3$, and check it at the zero pose against the figure.



2) Intermediate Transformations

Using the DH parameters below for the SCARA arm find the below homogeneous transformation matrices.

Scara Arm



Link	a_i	α_i	d_i	θ_i
1	a_1	0	0	$\theta_1^\star$
2	a_2	180	0	$\theta_2^\star$
3	0	0	$d_3^\star$	0
4	0	0	d_4	$\theta_4^\star$

* joint variable

(1) [2 points] H_1^0

(2) [2 points] H_2^1

(3) [2 points] H_3^2

(4) [2 points] H_4^3

3) Origin frames and Z axes

Calculate the Origin frames and Z axes at each joint.

(1) [2 points] Joint 1

(2) [2 points] Joint 2

(3) [2 points] Joint 3

(4) [2 points] Joint 4



3 Submission Checklist

- ☐ Upload `writeup.pdf` to Gradescope.